LESSON 2.8

Describe Data Using the Mean

Set B · Version A · Second form · supported practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

WHY THIS MATTERS Every week, Miguel walks dogs after school to earn money, and he keeps his earnings in a notebook. Across 4 days he earned $60 in all — but not the same amount each day. Later this lesson you'll help Sofia, who tracks her points every basketball game and wants to finish the season averaging 15 points per game.

REMEMBER

Describe Data Using the Mean

Measures of Center: Mean. Formula: Mean = (Sum of all values) ÷ (Total number of values) The mean, or average, is a measure of center that describes a whole data set with one number. It is what every value would be if the total were leveled out and shared equally.

1. Add all data values together to find the total sum.
2. Divide the sum by the count of data values.
3. Concept: The mean represents the fair share or balance point of the data

Words: **Mean *(Media)*** · **Average *(Promedio)*** · **Measure of center *(Medida de centro)*** · **Balance point *(Punto de equilibrio)*** · **Data set *(Conjunto de datos)***

1 **Circle the letter of the best answer. Show how you know.**

What is the range of 12, 7, 19, 4 and 15?

(A) 11 (B) 15

(C) 19 (D) 23

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

2 **Circle the letter of the best answer. Show how you know.**

A data set has Q1 = 8 and Q3 = 21. What is the interquartile range?

(A) 8
(B) 13
(C) 21
(D) 29

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Two classes have the same median. Class A has an IQR of 3 and Class B an IQR of 12. Whose scores are more spread out?

(A) Class B
(B) Class A
(C) They are the same
(D) You cannot tell from the IQR

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the range of 3, 3, 3, 3?

(A) 0
(B) 1
(C) 3
(D) 12

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

SENTENCE STARTER Counting games is not enough because ___. To balance, I have to measure how ___ each score is from ___, and then compare the total ___ to the total ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.8

Describe Data Using the Mean

Set B · Version B · Second form · core practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

WHY THIS MATTERS Every week, Miguel walks dogs after school to earn money, and he keeps his earnings in a notebook. Across 4 days he earned $60 in all — but not the same amount each day. Later this lesson you'll help Sofia, who tracks her points every basketball game and wants to finish the season averaging 15 points per game.

1 **Circle the letter of the best answer. Show how you know.**

Sofia tracks the number of points she scores in each basketball game. In games 1 through 9 she scored 12, 20, 16, 11, 17, 14, 15, 10, and 18 points. She wants to end the season with a mean of 15 points per game across all 10 games. Compared with that target of 15, her nine games came in 3 below, 5 above, 1 above, 4 below, 2 above, 1 below, even, 5 below, and 3 above. How many points has Sofia scored in games 1 through 9?

(A) 15 points

(B) 133 points

(C) 135 points

(D) 150 points

PER 1

2 **Circle the letter of the best answer. Show how you know.**

Sofia tracks the number of points she scores in each basketball game. In games 1 through 9 she scored 12, 20, 16, 11, 17, 14, 15, 10, and 18 points. She wants to end the season with a mean of 15 points per game across all 10 games. Compared with that target of 15, her nine games came in 3 below, 5 above, 1 above, 4 below, 2 above, 1 below, even, 5 below, and 3 above. To finish with a mean of 15 points across 10 games, how many points does Sofia need in all?

(A) 15 points
(B) 25 points
(C) 133 points
(D) 150 points

PER 1

3 **Circle the letter of the best answer. Show how you know.**

Sofia tracks the number of points she scores in each basketball game. In games 1 through 9 she scored 12, 20, 16, 11, 17, 14, 15, 10, and 18 points. She wants to end the season with a mean of 15 points per game across all 10 games. Compared with that target of 15, her nine games came in 3 below, 5 above, 1 above, 4 below, 2 above, 1 below, even, 5 below, and 3 above. Her nine games total 13 points below the target mean and 11 points above it. What must game 10 be for the data to balance at 15?

(A) 13 points — 2 below the target mean
(B) 15 points — exactly the target mean
(C) 17 points — 2 above the target mean
(D) 24 points — enough to double the points above

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Data set: 8, 12, 10, 14. What is the mean?

(A) 10
(B) 11
(C) 12
(D) 44

Put the data in order

WORK

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help